

Final Capstone Report

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1. Executive Summary

This report outlines the end-to-end data engineering and analytics pipeline built for the Indian Mutual Fund industry. Over INR 81 Lakh Crore of AUM data, 32,000+ investor transactions, and 40 top-performing schemes were ingested, cleaned, and modeled into a robust SQLite Star Schema.

2. ETL & Data Modeling Methodology

- Data Ingestion: Automated fetch of historical NAV, AUM, and portfolio CSVs.
- Data Cleaning: Addressed missing values using forward-fill, normalized transaction dates, and standardized fund naming conventions.
- Star Schema: Built around fact_nav, fact_transactions, and fact_performance, connected to dim_fund and dim_date.

3. Advanced Analytical Insights

1. Risk-Adjusted Dominance: Top tier funds maintained a Sharpe > 1.2 despite macro volatility.
2. Sectoral Tail Risk: Value-at-Risk (VaR) calculations exposed heavy tail risks in sectoral funds, corroborated by high HHI concentration.
3. SIP Stickiness: Cohort analysis proved that early adopters maintain active SIPs with a churn risk flagged at >35 days of inactivity.

4. Bonus Challenges Implemented

- Cron job automation for live NAV fetching.
- Monte Carlo simulations for 5-year NAV projections.
- Markowitz Efficient Frontier optimization.
- Interactive Streamlit Web Dashboard deployment.
- Automated HTML Email Report generation.